# openKVM v1 — KVM-over-IP ESP32-P4

Schema preliminaire rev 0.1 — revue humaine obligatoire avant fabrication.

Chaque net est porte par une etiquette globale : memes noms = meme connexion.

Alimentation

USB-C, OU ideal 5 V, buck 3,3 V, buck 1,2 V

File: power.kicad_sch

ESP32-P4

SoC, flash, quartz, DCDC coeur, bouton, LED, debug

File: mcu.kicad_sch

Capture HDMI

Connecteur HDMI, ESD, TC358743, oscillateur 27 MHz

File: hdmi.kicad_sch

Ethernet

PHY IP101GRI, quartz 25 MHz, RJ45 a magnetiques integres

File: eth.kicad_sch

Peripheriques

microSD (UHS-I), ESD USB

File: periph.kicad_sch

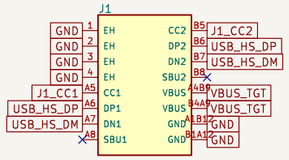
Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
**openKVM — CERN-OHL-S-2.0**

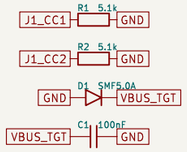
Sheet: /
File: openkvm.kicad_sch

***Title: openKVM v1 — vue d ensemble***

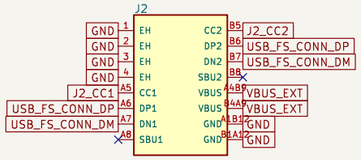
| Size: A4 | Date: 2026-11-12 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/6 |

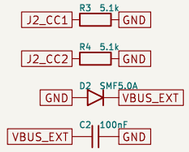
## USB-C CIBLE



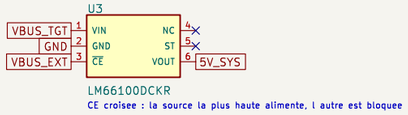




## USB-C EXT

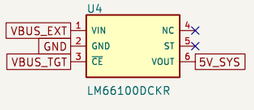


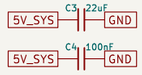




## OU ideal 5 V



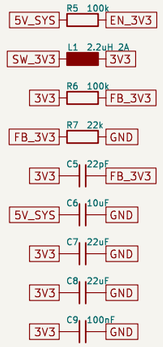

CE croisee : la source la plus haute alimente, l autre est bloquee



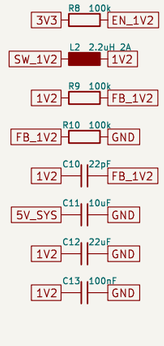

## Buck 3,3 V




## Buck 1,2 V (TC358743)



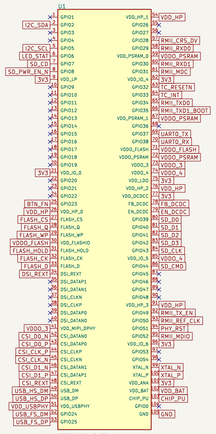

Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
**openKVM — CERN-OHL-S-2.0**

Sheet: /Alimentation/
File: power.kicad_sch

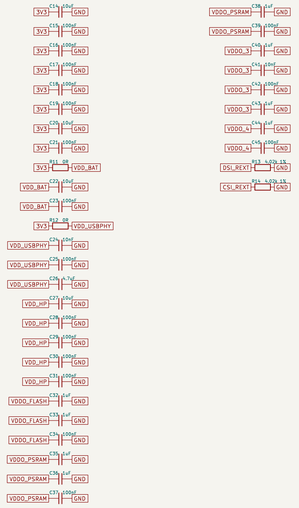
***Title: openKVM — Alimentation***

| Size: A3 | Date: 2026-11-12 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/6 |

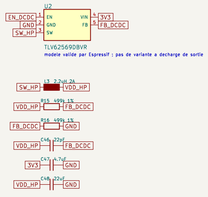
## ESP32-P4




## Decouplage P4 (au plus pres des broches)



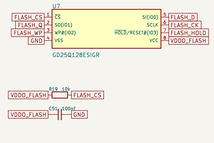
## DCDC coeur (pilote par le P4)




## Quartz 40 MHz

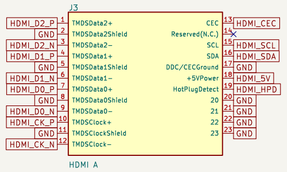


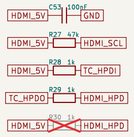

## Flash SPI 16 Mo

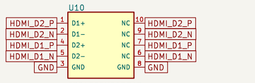



## Reset, boot, bouton, LED

## Connecteur HDMI



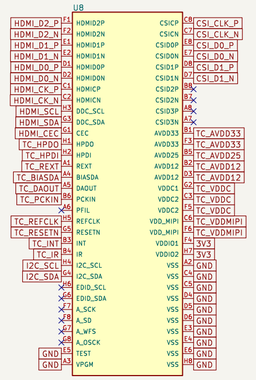



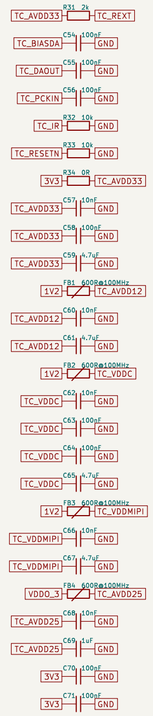




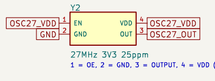


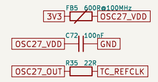

## TC358743XBG




## Oscillateur 27 MHz

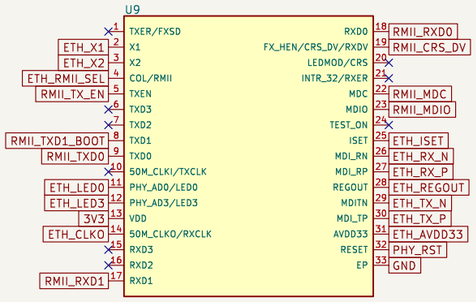


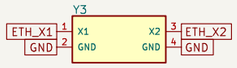



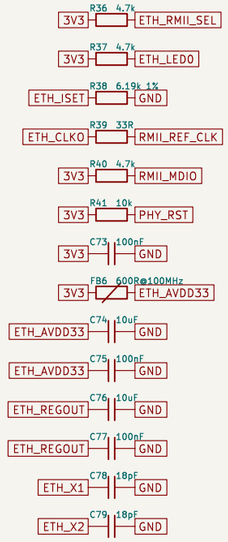


Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
**openKVM — CERN-OHL-S-2.0**
Sheet: /Capture HDMI/
File: hdmi.kicad_sch
***Title: openKVM — Capture HDMI***
Size: A2 | Date: 2026-11-12 | **Rev: 0.1**
KiCad E.D.A. 9.0.9 | Id: 4/6

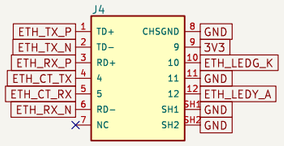
## PHY IP101GRI

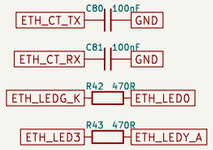



## RJ45



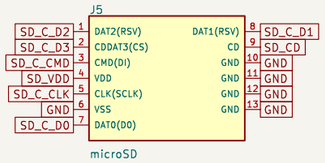

Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
**openKVM — CERN-OHL-S-2.0**

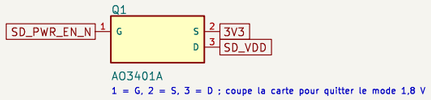
Sheet: /Ethernet/
File: eth.kicad_sch

***Title: openKVM — Ethernet***

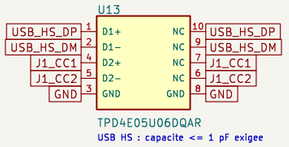
| Size: A3 | Date: 2026-11-12 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 5/6 |

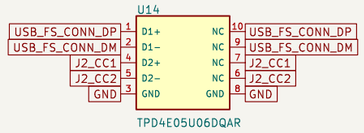
## microSD (slot SDMMC 0, UHS-I)

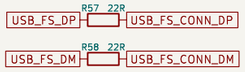


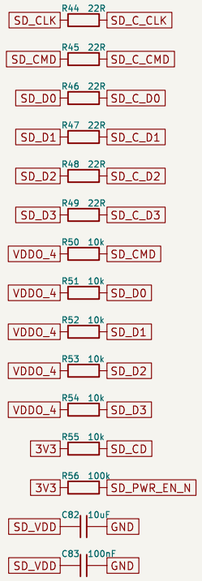




## ESD USB




Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
**openKVM — CERN-OHL-S-2.0**

Sheet: /Peripheriques/
File: periph.kicad_sch

***Title: openKVM — Peripheriques***

| Size: A3 | Date: 2026-11-12 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 6/6 |